MIL-S-777B(MU)
28 January 1972
SUPERSEDING
MIL-S-777A(Ord)
24 June 1958

MILITARY SPECIFICATION

SIGHT, REFLEX: M24C

1. SCOPE

1.1 This specification covers one type of reflex sight known as Sight, Reflex: M24C.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part pf this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components of Fire Control Instruments; General Specification Governing the Manufacture, Assembly and Inspection of
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MTL-STD-1O9	Quality Assurance Terms and Definitions

DRAWINGS

US Army, Frankford Arsenal

F7648288	Sight, Reflex: M24C
D8204694	Final Inspection Fixture Sight, Reflex: M24C

PACKAGING DATA SHEET

7648288	Packaging of Sight, Reflex: M24C

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(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.2 Fabrication.- The reflex sight, shall be fabricated in accordance with Drawing F7648288 and drawings pertaining thereto.

3.3 General specifications.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of MIL-O-13830 and MIL-F-13926.

3.3.1 MIL-O-13830

(a) Prisms and mirrors
(b) Lenses

3.3.2 MIL-F-13926

(a) Order of Precedence
(b) Dimensions and Tolerances
(c) Inorganic Protective Surface Finishes
(d) Part Identification and Marking
(e) Workmanship

3.4 Environmental.

3.4.1 Storage at extreme temperatures.- The sight shall not evidence damage after storage at -80° Fahrenheit (F) and +160°F.

3.4.2 Rain.- The sight shall show no evidence of water penetration after exposure to a simulated rainfall of 4 to 5 inches per hour for a period of two (2) hours.

3.5 Performance.

3.5.1 Projected reticle pattern definition.- The projected reticle pattern image lines shall show no breaks when viewed by reflected light from the partial reflector.

3.5.2 Open sight.- The line of sight of the open sight, when thrown into operating position shall be parallel to the line of sight through the partial reflector within three (3) degrees as measured at the midpoint of the reticle image.

3.5.3 Focus.- The reticle image shall focus within 0.25 diopter of the target focus point when viewing a 1000±200 meter target along a line

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of sight parallel to the open sight within three degrees (3″) which passes through the partial reflector 1.25±0.25 inch above the top surface of Body Assembly 7642128. This requirement shall also be met at four angular position 90°±5° apart on a 0.75±0.06 inch radius circle concentric to the line of sight.

3.5.4 Focus shift.- The focus shift between vertical and horizontal reticle image lines shall not exceed 0.25 diopter when viewing along the line of sight defined in 3.5.3.

3.5.5 Reticle position.- With the mounting axis of the sight in a vertical position, the image of the vertical reticle line shall be vertical within an angle subtended by a two (2) mil block at one end of the reticle line while the other end is in coincident with the respective target reference line.

3.5.6 Optical component defects.- There shall be no visual evidence of fracture or adhesive separation on optical components of the sight.

3.5.7 Optical component cleanliness.- There shall be no visual evidence of grease, fingerprints, or condensates on optical elements. The partial reflector and objective lens top surface shall not exhibit more than three particals of dirt or foreign matter (classified as dirt), and the size of any visible particle shall not exceed 0.1 inch.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or Purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities Or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-O-13830 and MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be listed in MIL-STD-109.

4.2 First article (initial production) approval. The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) sights plus three (3) each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Govvernmet, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

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4.2.2 Contractor testing. - When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAp) listed in the technical data package (TDP). In the absence of SQAPS, the applicable Quality Assurance Provisions of MIL-O-13830 and MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection.- Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I and 11 herein. Examination and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I and II shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS

Class		REQUIREMENTS	TEST PROCEDURES
CRITICAL: NONE			
MAJOR: AQL 1.0 percent defective			
101.	Projected reticle pattern definition	3.5.1	4.6.1
102.	Open sight	3.5.2	4.6.2
103.	Focus	3.5.3	4.6.4
104.	FOCUS shift	3.5.4	4.6.4
105.	Reticle position	3.5.5	4.6.3
106.	Optical component defects	3.5.6	4.6.7
107.	Optical component cleanliness	3.5.7	4.6.7
MINOR: NONE			

NOTE: The tests in Table I shall be conducted in accordance with the test condition requirements in the "Test Facilities" requirement of MIL-F-13926.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units

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and resubmittance of rejected lots shall be in accordance with “Acceptance and Rejection" as specified in MIL-STD-105.

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4.4 Control tests.

4.4.1 Special sampling.- Samples shall be selected from lots at random using inspection level S-3 Table I of MIL-STD-105. Acceptance or rejection of the lot shall be in accordance with AQL 2.5 of Table II-A of MIL-STD-105. The samples shall have met all the requirements and tests outlined in Table I prior to and after being subjected to the following examinations and tests,

TABLE II

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301. Storage at extreme temperature	3.4.1	4.6.5

4.4.1.1 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements. acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.4.2 First article sample.- Unless otherwise specified in the contract. test(s) listed herein shall be conducted on one unit of the initial production sample and shall not again be performed on the remaining production quantities, provided the manufacturing processes and techniques used to produce the samples have not been changed or revised.

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
302. Rain	3.4.3	4.6.6

4.4.3 Certification.- Certification shall include test data and results for all characteristics specified in Table III. Certification shall be provided prior to performing final acceptance inspection, and remain in effect throughout the contract for Government acceptance. This does not relieve the contractor from inspection throughout the contract production, including recording data and results therefrom. Recorded data and results shall be made available to the Government upon request. A new certification will be required should any changes occur in materials, manufacturing processes and techniques previously certified.

TABLE III

CHARACTERISTIC	REQUIREMENT	TEST DATA TO COMPLY WITH
Materials	3.1	Applicable drawings Standards, & Specifications

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TABLE III (cont'd)

CHARACTERISTIC	REQUIREMENT	TEST DATA TO COMPLY WITH
Fabrication	3.2	Standards, Specifications and applicable Drawings
General specification (MIL-F-13926)	3.3.2	
General specification (MIL-O-13830)	3.3.1	

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable-requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specification or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the Technical Data Package (TDP),

4.5.2.2 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 and with the "Component Inspection" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.- The test equipment provided in accordance with 4.5.2.2 shall meet the following requirements:

(a) Target.- Figure 1 is used in conjunction with Fixture 8204694 and a viewing collimating telescope to facilitate checking the requirements of paragraphs 3.5.2 and 3.5.5; and when in use, the target is placed 12 foot, 6 inches from the collimator objective lens when the collimator is placed in the fixture. The target is used to determine either the horizontal and/or vertical positioning of the reticle line to their respective means for mounting and also for checking parallelism of the open sight with respect to the reticle. The target contains mil blocks (E) at each end of the Vertical Line "C". The size of a mil block for these targets (for all practical purposes) can be determined by

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dividing the target distance (in inches) by 1000. Target Lines “A” and “B” are provided for aligning the target to the horizontal and vertical reticle lines of the collimator, when the collimator is positioned in the test fixture and when the collimator’s vertical line of sight is in coincidence with a plumb line. The location of Lines “A” and “B”, with respect to Lines “C” and “D”, will be constant for any target distance. The length and thickness of lines "C" and "D" will vary in proportion. to the target distance, and will be calculated from the following formula:

$$D = \frac{HS}{F}$$

D = The distance on the target from the geometric axis to any line or mark on the target.
H = The distance from the geometric axis of the reticle to the corresponding line or mark on the reticle.
S = The distance from the objective being tested to the target in inches.
F = The equivalent focal length of the reflex sight’s objective lens.

(b) Alignment of the target.- prior to aligning the target, level fixture 8204694 on the surface plate by means of a bench level. The fixture is considered level when the bubble of the level vial maintains a constant position when the second position of level is placed 90° to the first position. place the collimator into the fixture’s “V” slot and direct its line of sight toward a plumb line approximately 12 1/2 feet from the fixture. Coincide its vertical line to the plumb line. This coincidence is obtained by rolling the collimator in the “V” slot and/or by shifting the fixture on the surface plate. When coincedence has been established, place Target Line “A” in coincidence with the plumb line and collimator’s vertical reticle line. Without disturbing the collimator or the coincidence. mentioned above, raise or lower the target until target line “B” is in coincidence with the collimator’s horizontal reticle line. When using the collimating telescope at this distance a peep sight must be used.

4.6 Test methods and procedures.

4.6.1 Projected reticle pattern definition.- This test shall be performed visually using any 3-volt light to illuminate the reticle, mounted at the reticle window. The image of the reticle, as viewed on the reflectors, shall be as specified in 3.5.1.

4.6.2 Open sight.- Position the open sight as specified in 3.5.2, and sight through the open sight toward the Target Point “P” to determine that the open sight falls within the tolerance as specified in 3.5.2. The circle formed by Radius “R” is provided to facilitate this check. The equipment shall be capable of inspecting the open sight within 0° 18’.

4.6.3 Reticle position. - Remove the collimator and position the reflex sight in the fixture, sufficiently clamped, to avoid shifting. Illuminate the reflex sight reticle with an instrument light at window 7648310 and sight through the reflector and observe coincidence between the reticle line and Target Line “C”. Should either end of the reticle line

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be in coincidence with its respective target line, then the opposite end of this reticle line must coincide with the opposite end of the target line within the tolerance specified in 3.5.5 when viewed at the proper eye distances. This tolerance is shown on the target as Mil Block "E" of Figure 1. The equipment shall be capable of inspecting the optical and mechanical axis of the reflex sight within 0.1 mil.

4.6.4 Focus and focus shift.- Establish the line of sight defined in 3.5.3 and utilize a dioptometer whose entrance aperture is located eight to ten inches from the vertical centerline of Objective 7648295. Record the point at which the target and reticle image can be focused along the line of sight and at four positions specified in 3.5.4. Verify compliance to 3.5.3 and 3.5.4. When conducting this test, the target or sight may be moved angularly in azimuth or elevation as required to bring about alignment adequate to measure focus shift.

4.6.5 Temperature.- The test equipment to be utilized when performing the temperature test shall be such that the equipment under test shall not exceed 50 percent of the internal volume of the test chamber and no portion shall be closer than 6 inches from any surface of the chamber. Plugs, covers, and inspection plates used in service shall remain in place. When electrical or mechanical connections are not utilized, the connections normally covered in service shall be adequately covered. The equipment shall be installed in the chamber in manner similar to its actual use in service. The reflex sights shall be tested in accordance with Procedure 1, Temperature Cycling in Specification NIL-F-13926 to meet the requirements as specified in 3.4.1.

4.6.6 Rain.- The equipment shall be mounted in a test chamber to simulate installation conditions. The rain test chamber shall be capable of maintaining a temperature of between 20°C to 30°C (68 °F to 86°F) throughout the test period. A simulated rainfall as measured at the surface of the equipment with a U.S. Weather Bureau type gage shall be produced by means of a water spray nozzle of such design that the water is emitted in the form of small droplets rather than a fine mist. The temperature of the water shall be maintained between 11°C to 20°C (52°F to 68°F. The rainfall shall be dispersed uniformally over the test area within the limits as specified above. Direction of rainfall shall be variable from vertical to 45 ° from vertical in any direction. The simulated rainfall shall be directed on all surfaces which are normally subjected to rain and shall meet the requirements as specified in 3.4.2.

4.6.7 Optical component defects and cleanliness. - Subsequent to conducting all tests and examinations and without sight dis-assembly, perform a visual examination of all optical elements and areas between elements to verify compliance to 3.5.6 and 3.5.7. This test shall be performed by viewing in an illuminated area such as would exist by holding the sight in close proximity to a bench lamp containing two fifteen watt daylight fluorescent tubes backed by a white reflector.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with the applicable Packaging Data Sheet specified

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in Section 2. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The Reflex, Sight: M24C is designed for use as a tracking aid for direct fire against aircraft.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification
b. Requirement for First Article Testing
c. Applicable packaging data sheet number (see 5.1)
d. Selection of applicable levels of preservation, packaging and packing

Custodian:

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Preparing activity:

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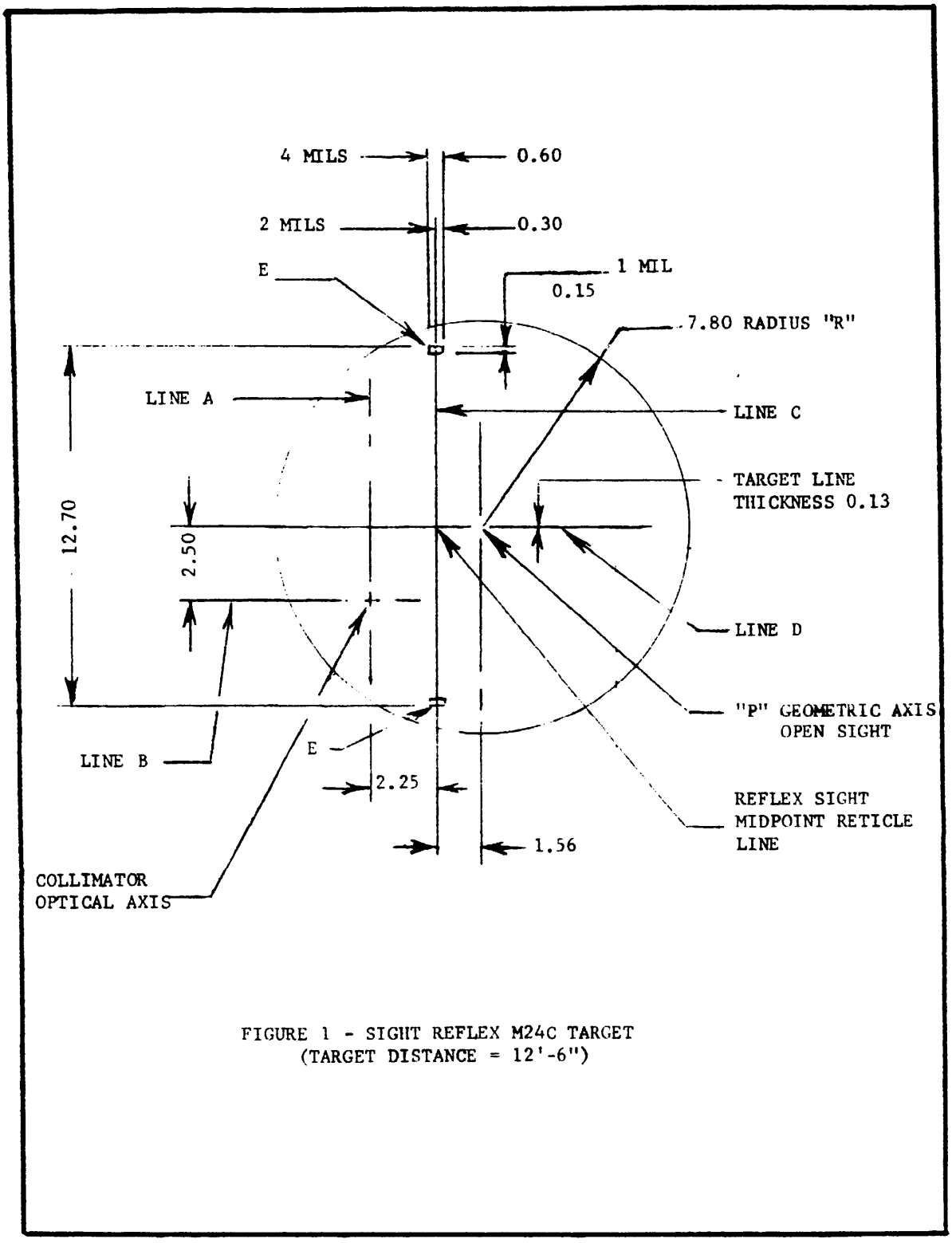


FIGURE 1 - SIGHT REFLEX M24C TARGET
(TARGET DISTANCE = 12'-6")

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426

REPLACES EDITION OF OCT 64 WHICH MAY BE USED

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